

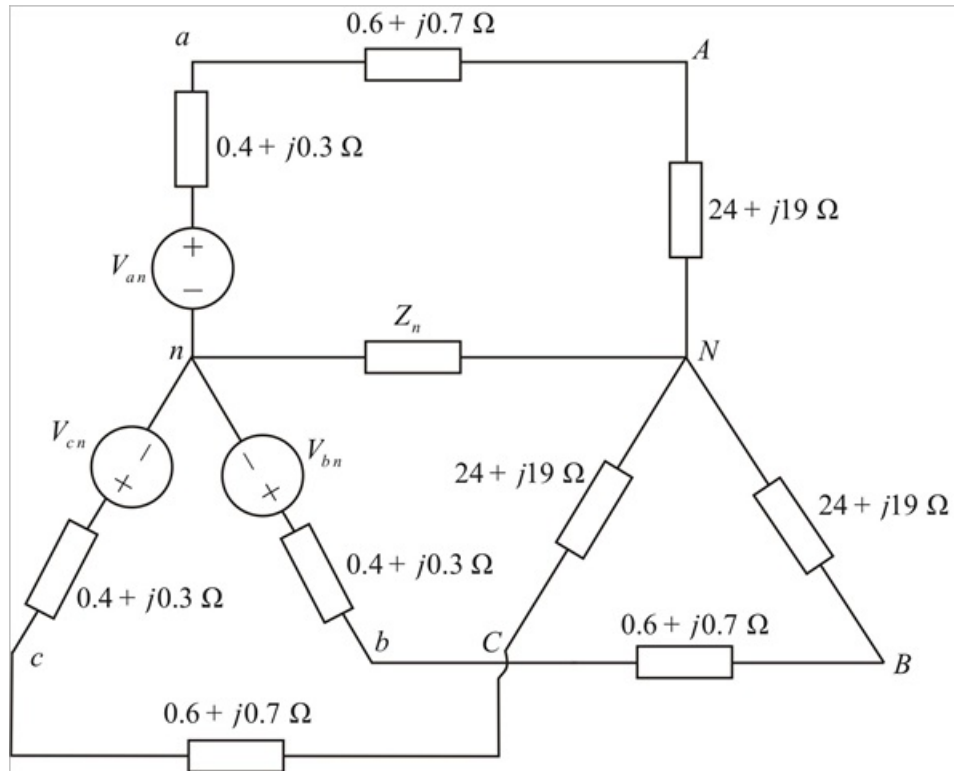
## Problem

For the Y-Y circuit in Practice Prob. 12.2, calculate the complex power at the source and at the load.

## Step-by-step solution

## Step 1 of 4

According to given values the  $Y-Y$  balanced system is



## Step 2 of 4

Given that

$$V_{an} = 120 \angle 30^\circ \text{ V}$$

Therefore

$$V_{bn} = 120 \angle 30^\circ - 120$$

$$V_{bn} = 120 \angle -90^\circ \text{ V}$$

$$V_{cn} = 120 \angle 30^\circ - 240$$

$$V_{cn} = 120 \angle -210^\circ \text{ V}$$

## Step 3 of 4

The phase voltage is

$$V_p = V_{an}$$

$$V_p = 120\angle 30^\circ \text{ V}$$

The phase current is,

$$I_p = \frac{V_{an}}{Z_Y}$$

Where  $Z_Y = 0.4 + j0.3 + 24 + j19 + 0.6 + j0.7$

$$Z_Y = 25 + j20$$

$$Z_Y = 32\angle 38.66^\circ \Omega$$

Hence

$$I_p = \frac{120\angle 30^\circ}{32\angle 38.66^\circ}$$

$$\therefore I_p = 3.75\angle -8.66^\circ \text{ A}$$

#### Step 4 of 4

Thus, at the source the complex power absorbed is

$$S_s = -3V_p I_p^*$$

$$S_s = -3 \times (120\angle 30^\circ) \times (3.75\angle 8.66^\circ)$$

$$\therefore \boxed{S_s = -(1054 + j843.3) \text{ VA}}$$

At the load the complex power absorbed is

$$S_L = 3|I_p|^2 Z_p$$

$$S_L = 3 \times 3.75^2 \times (24 + j19)$$

$$\therefore \boxed{S_L = 1012.5 + j801.6 \text{ VA}}$$